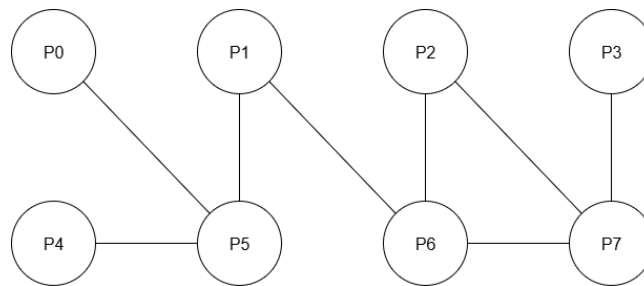


Assignment Sheet

Task 1 – Graph Structures

(3 Points)

Decide the diameter, degree and number of connections for the following graph structure G:



Diameter(G)=

Degree(G)=

|Connections|(G)=

Further Explanations:

Task 2 – Mesh

(5 Points)

Decide diameter, degree and number of connections for a mesh structure M with nodes, $P_{x,y}$ where all nodes with the same value of x and the same value of y form a ring. Maximum values for x and y are defined as X and, Y respectively, meaning that $0 \leq x \leq X - 1$ and $0 \leq y \leq Y - 1$. You may assume that both X and Y are even values. Express them in terms of X and Y if applicable.

Diameter(M)=

Degree(M)=

|Connections|(M)=

Further Explanations:

Task 3 – GPU's

(3 Points)

Explain why GPU's traditionally fall into the category of SIMD systems.

Your answer:

Task 4 – MESI Protocol

(9 Points)

Complete the following table with the correct MESI-Protocol states for Processors P1, P2, and P3 for data entries X and Y.

You may use the shorthand forms X:M, X:E, X:S and X:I denoting that the value of X for a given processor is modified, exclusive, shared, or invalid, respectively, same with the value Y:

Action	P1	P2	P3
Initial			
P1 reads X			
P2 reads X			
P2 writes Y			
P3 reads Y			
P1 writes X			
P2 reads X			
P3 writes Y			
P1 reads Y			

This assignment is due on 16.01.2026 at 10 am.